1.Diabetes Data Analysis

1.1 Describe the relationship between variables

Answer: A correlation matrix computed to examine linear relationships among the variables. The heatmap visualization shows strong correlations among the blood serum measurements (S1, S2, S3, S4, S5, and S6). BMI and BP exhibit moderate correlation, while AGE and SEX have relatively weak correlations with most variables.

1.2. What is collinearity? What effect does collinearity amongst predictor features/ variables have on their estimated coefficient value?

Answer:

Collinearity this occur when two or more predictor variable are highly correlated

Effects on coefficient estimate are

- Regression coefficients become unstable
- Standard errors increase
- Variables may appear statistically insignificant even when important
- Model interpretation becomes unreliable

1.3. Are all variables significant? Could this be a problem of collinearity?

Answer:

- A multiple linear regression model was fitted using all ten explanatory variables and a constant term. Model performance was evaluated using Mean Squared Error (MSE) and Adjusted R^2. Not all variables are statistically significant (p-value > 0.05). This lack of significance is likely due to collinearity, especially among serum variables.
- Yes, collinearity is a problem in this model

1.4. What is the difference between forward selection and backward selection?

Answer:

The differences between forward and backward election are; -

Forward selection begins with no predictors and adds variables sequentially based on improvement in model performance**.**

WHILE

Backward selection starts with all predictors and removes the least significant variables iteratively.

1.5. How does the approach stepwise work in the sense of selecting variables? Use the stepwise function to interactively compose a model using forward selection. Which variables are selected? How does this function work? What is the MSE and R2 value for this new model?

Answers:

The way stepwise function work is as follow:

- The process starts with an empty model.
- Each predictor variable is tested individually by adding it to the current model.
- The variable that produces the greatest improvement in model performance.
- This variable is permanently added to the model.
- The procedure repeats by testing the remaining variables.
- The process stops when no additional variable.

Variables are selected are

- BMI, BP, S5, AGE

2. Analyzing the Titanic data set

2.1. What is the difference between logistic regression and linear regression?

Answer:

Linear regression is used when the dependent variable is continuous (numerical). It models the relationship between the predictors and the outcome by fitting a straight line that minimizes the squared differences between observed and predicted values.

WHILE

Logistic regression is used when the dependent variable is binary or categorical (e.g., 0/1, Yes/No, Survived/Not Survived). Instead of predicting a continuous value, it estimates the probability that an event occurs.

2.4. Build a logistic regression model for survival rates based on passenger class, sex and age. What are the parameter estimates and are these parameters statistically significant?

Answer:

- A logistic regression model was fitted using passenger class, sex, and age as predictors. The estimated coefficients indicate that **gender and passenger class** are statistically significant predictors of survival**.**

3. PCA

3.1 Give a qualitative description of Principal Component Analysis (PCA) and its applications in machine learning. Why might it be useful to consider PCA to transform a set of explanatory variables?

Principal Component Analysis (PCA)

- Is a dimensionality reduction technique used in statistics and machine learning. Its main goal is to transform a set of possibly correlated variables into a smaller set of uncorrelated variables, called principal components, while retaining as much of the original data's variability (information) as possible.

Its application in machine learning

- Dimensionality Reduction: Reduces the number of features while keeping most of the information.
- Noise Reduction: PCA can filter out components with low variance, which often correspond to noise.
- Visualization: Helps in projecting high-dimensional data into 2D or 3D for plotting and exploring patterns.

PCA is useful for transforming explanatory variable

- Reduces Multicollinearity: When explanatory variables are highly correlated, regression models can become unstable.
- Simplifies the Model: Using a few principal components instead of many variables can make the model simpler and faster to train.
- Focuses on Most Important Information: PCA prioritizes components that capture the largest variance, which often contains the most relevant signal in the data.
- Facilitates Visualization & Interpretation: Transforming complex datasets into principal components allows easier identification of patterns and relationships.

3.2 Write down the mathematical equations for PCA explaining how one transforms the raw input data matrix X into a new set of variables. Give an interpretation of each matrix.

Mathematic equations for PCA

let

$$X = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1p} \\ X_{12} & X_{22} & \dots & X_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ X_{n1} & X_{n2} & \dots & X_{np} \end{bmatrix}_{n \times p}$$

Where

n - number of observation (rows)

p - number of variables (column)

Step 1: Standardize the data

PCA works better if variables are standardize (mean = 0, variance = 1);

$$Z = \frac{X - \mu}{\sigma}$$

Where

X = Original data matrix $(n \times p)$

μ = vector of means for each variable $(1 \times p)$

σ = vector of standard deviations for each variable $(1 \times p)$

Z = standardized data matrix

Step 2: Compute the covariance matrix

$$\Sigma = \frac{1}{n-1} Z^T Z$$

Step 3: Eigen decomposition

$$\Sigma v_k = \lambda_k v_k, \quad k = 1, 2, 3 \dots, p$$

Where

λ_k - eigenvalue for principal component k

v_k - eigenvalue (length p)

Step 4: Compute principal components

$$Y = Z V$$

Where

Y - $(PC1, PC2, \dots, PCp)$ is the transformed data $(n \times p)$

V - $(v_1, v_2, \dots v_p)$ is the matrix of eigenvalue $(p \times p)$.

PC_k - ~~Zv_k~~ $Z v_k$ is the principal component